# RISHANT PRAKASH

Ph.D. student in Planetary and Space Sciences | Department of Physics | University of Central Florida

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**Research interests:** Cosmochemistry; Solar System origins and volatile depletion; high- and low-temperature geochemistry; isotope measurements and mass spectrometry.

## EDUCATION

**University of Central Florida** 2026-Present
*Ph.D. in Planetary and Space Sciences | Department of Physics*

**Arizona State University | School of Earth and Space Exploration** 2023-2026
*M.S. in Geological Sciences; GPA: 3.9/4.0*

**Indian Institute of Science, Bengaluru** 2021-2023
*M.Tech. in Earth Sciences; GPA: 7.9/10*

**Hansraj College, University of Delhi** 2019-2021
*M.Sc. in Geology; GPA: 8.32/10; First Division with Distinction*

**Hansraj College, University of Delhi** 2016-2019
*B.Sc. in Geology; GPA: 7.46/10*

## RESEARCH EXPERIENCE

**Titanium isotope anomalies in the solar nebula** Started Aug 2023
*Previous research, School of Earth and Space Exploration, Arizona State University*

**Advisor:** Prof. Meenakshi Wadhwa

- Developed and refined in-situ LA-MC-ICP-MS protocols for high-precision Ti isotope measurements of diverse calcium-aluminum-rich inclusions (CAIs), including FUN CAIs.
- Synthesized Ti-doped CMAS glasses with varied V and Cr ratios as internal standards, using a gas-mixing furnace at the University of Arizona.
- Performed Mg purification in achondrites, including column chemistry, acid titration, and hazardous-waste management in Class 1000 cleanrooms.

**Presolar grains and core-collapse supernovae** Started Aug 2024
*Previous research, School of Earth and Space Exploration, Arizona State University*

**Advisors:** Prof. Patrick Young and Prof. Larry Nittler

- Used three-dimensional core-collapse supernova models to investigate Ti and Cr isotopic signatures measured in presolar oxide grains.
- Examined explosive nucleosynthesis, mixing, and fallback; compared simulated signatures with laboratory NanoSIMS datasets to investigate astrophysical source reservoirs.

**Acid leaching, silicate weathering, and meteorite geochemistry** 2021-2023
*M.Tech. research, Centre for Earth Sciences, Indian Institute of Science, Bengaluru*

**Advisor:** Prof. Ramananda Chakrabarti

- Conducted acid-leaching experiments on ultramafic to felsic silicate rocks, using non-HF acids and high-pressure asher digestion to investigate cation mobility.
- Examined silica content, residue weight, mineral stability, and major- and rare-earth-element behavior in leachates and residues.
- Characterized the Sadiya meteorite using bulk Ni, Co, and Fe data to investigate its H/L-chondrite classification.

Thesis

## RESEARCH EXPERIENCE - CONTINUED

**Amoeboid olivine aggregates in CM and CV chondrites** May-Jul 2022

*Summer research internship, Planetary Science Division, Physical Research Laboratory, Ahmedabad*

**Advisor:** Dr. Dipak Panda

- Analyzed amoeboid olivine aggregates in Mukundpura and Allende chondrites using EDS and WDS on a JEOL 8530F field-emission electron-probe microanalyzer.
- Compared mineral chemistry and alteration histories, documenting phyllosilicates, carbonates, and poorly characterized phases in Mukundpura.
- Examined low-Ca pyroxene at aggregate margins to investigate nebular condensation and links with chondrule formation; presented the work at MetMESS 2022.

Project report

**Land-use and land-cover change in Dehradun** 2020-2021

*M.Sc. dissertation, Department of Geology, University of Delhi*

**Advisor:** Prof. Vimal Singh

- Used Landsat 7 ETM, Landsat 4-5 TM, and Landsat 8 OLI/TIRS imagery to classify land use and cover in 2001, 2011, and 2021.
- Applied maximum-likelihood supervised classification in ERDAS IMAGINE; assessed accuracy using error matrices and Kappa statistics and interpreted spatial and temporal changes.

Dissertation

**Depositional environments in the Krishna-Godavari Basin** 2019

*Industry internship, Oil and Natural Gas Corporation, Dehradun*

**Advisor:** GM Rajesh Sharma

- Integrated gamma-ray and resistivity logs, core observations, and biostratigraphy from four wells to interpret fluvial to marginal-marine synrift sequences.
- Analyzed sandstone maturity and diagenesis to assess reservoir quality and fluid mobility; used biostratigraphic and palynological evidence to establish Late Jurassic to Early Cretaceous ages.

Internship report

## RESEARCH CONTRIBUTIONS

**Prakash, R.**, Young, P. A., and Desch, S. J. *Three-dimensional core-collapse supernova clumps as sources of $^{54}$Cr-rich presolar oxide nanograins.* 88th Annual Meeting of the Meteoritical Society, Frankfurt am Main, LPI Contribution 3162, abstract 5283. **Conference abstract; oral session.** Abstract 2026

**Prakash, R.**, Putak Juricek, M., Spring, I., Rai, V., and Wadhwa, M. *Development of V- and Cr-doped Ti-CMAS glasses as standards for in situ titanium isotope analysis of calcium-aluminum-rich inclusions.* 88th Annual Meeting of the Meteoritical Society, Frankfurt am Main, LPI Contribution 3162, abstract 5278. **Conference abstract; poster session.** Abstract 2026

**Prakash, R.** and Young, P. *Presolar oxide grains as probes of symmetric and asymmetric core-collapse supernova explosions.* 247th American Astronomical Society Meeting, January 4-8, 2026, abstract 277.01. **Conference abstract.** Abstract 2026

**Prakash, R.** *Thermodynamic origins of $^{54}$Cr-rich presolar oxide nanograins from three-dimensional core-collapse supernova models, with synthesis and characterization of Ti-CMAS glasses for in situ titanium isotope analysis.* Arizona State University. **Thesis.** Record 2026

Campbell, A., **Prakash, R.**, Chakraborty, A., Vierling, S., Garvie, L., and Wittmann, A. *Santa María del Oro (IIIAB, iron), Durango, Mexico.* Meteoritical Bulletin, No. 114; approved July 2025. **Classification contribution.** Database entry 2025

**Prakash, R.**, Rai, V. K., Aigner, M., Hines, R., and Wadhwa, M. *Origin and distribution of Ti nucleosynthetic anomalies in the solar nebula.* SESE Symposium, February 22, 2025. **Poster.** 2025

**Prakash, R.** and Panda, D. *Geochemical analysis of amoeboid olivine aggregates in Mukundpura and Allende chondrites.* MetMESS, November 24-25, 2022, abstract S1-09. **Conference abstract and presentation.** Abstract 2022

## AWARDS AND SCHOLASTIC ACHIEVEMENTS

- NASA Travel Award, 88th Meteoritical Society Annual Meeting, Frankfurt, Germany 2026
- SESE Exploration Graduate Fellowship, Arizona State University 2026
- Graduate College Travel Award, 247th American Astronomical Society Meeting 2026
- University Graduate Fellowship, College of Liberal Arts and Sciences, ASU 2025
- GATE Scholarship for M.Tech. study, Indian Institute of Science 2021-2023
- Summer Research Fellowship, Physical Research Laboratory, Ahmedabad 2022
- M.Sc. First Division with Distinction, Hansraj College, University of Delhi 2021
- Selected student volunteer, International Geological Congress 2020
- Second place, National Geological Quiz, AAPG Student Chapter, University of Delhi 2016-2017

## GEOLOGICAL FIELD TRAINING

**Meteor Crater and Sunset Crater, Arizona, USA** Sep 2023
Studied overturned strata and impact-crater formation at Meteor Crater, aa lava flows at Sunset Crater, and distinctions between impact and fluid-interaction crater processes.

**Ajeetgarh, Rajasthan, India** Dec 2019
Mapped Precambrian geology and structural deformation; examined foliation, lineation, mineralization, and cross-bedding in the basal Alwar Group.

**Dehradun-Mussoorie, Uttarakhand, India** Sep 2018
Studied structural and sedimentological features, the Himalayan Frontal and Main Boundary thrusts, and the Ganga-Yamuna drainage divide; practiced geological mapping and field instruments.

**Varanasi and the Vindhyan region, Uttar Pradesh, India** Mar 2018
Collected Semri and Rewa formation samples, examined Proterozoic sedimentary structures, and measured stromatolites in the Sulkan Valley.

**Nainital, Uttarakhand, India** Dec 2016
Examined the geological setting of Nainital Lake and the structural and sedimentological characteristics of the Krol and Tal formations.

## OUTREACH, LEADERSHIP, AND SERVICE

**SESE Outreach, Buseck Center for Meteorite Studies, ASU** From 2023
Shared extraterrestrial and terrestrial rocks with visitors from varied backgrounds as part of previous ASU outreach.

**IISc Open Day, Centre for Earth Sciences** 2022-2023
Introduced visitors to meteorites and science experiments.

**Coordinator, AAPG Student Chapter, University of Delhi** 2018-2019
Organized events and quiz competitions and managed chapter social media.

**Technical Head, PIXELS Photography Society, Hansraj College** 2017-2019
Led 40 undergraduates; organized photo walks, taught photojournalism and editing, and documented campus events.

**Volunteer Teacher, NAV UMEED, North Delhi** 2017-2019
Supported high-school students from marginalized backgrounds with mathematics and science in evening classes.

**Technical Head, VISION Newspaper Society, Hansraj College** 2017-2018
Led editorial design and publication of a biweekly student newspaper.

**Volunteer, National Service Scheme Rural Development Wing** 2016-2018
Facilitated workshops on personal hygiene, education, and poverty alleviation for marginalized youth in North Delhi.

## TECHNICAL SKILLS

**Analytical instruments:** LA-MC-ICP-MS (Neptune; Photon Machine 193 nm UV laser); ICP-MS (Thermo X-Series II); EPMA (JEOL); SEM (Phenom XL).

**Laboratory and experimental:** Class 1000 and 10000 cleanrooms; column chemistry; acid digestion; gas-mixing furnace; high-pressure asher.

**Programming:** Python, C++, LaTeX.

**Software:** ArcGIS, ERDAS IMAGINE, OriginPro, Google Earth, Adobe Suite, Microsoft Office, Google Suite.

**Languages:** English and Hindi (fluent).